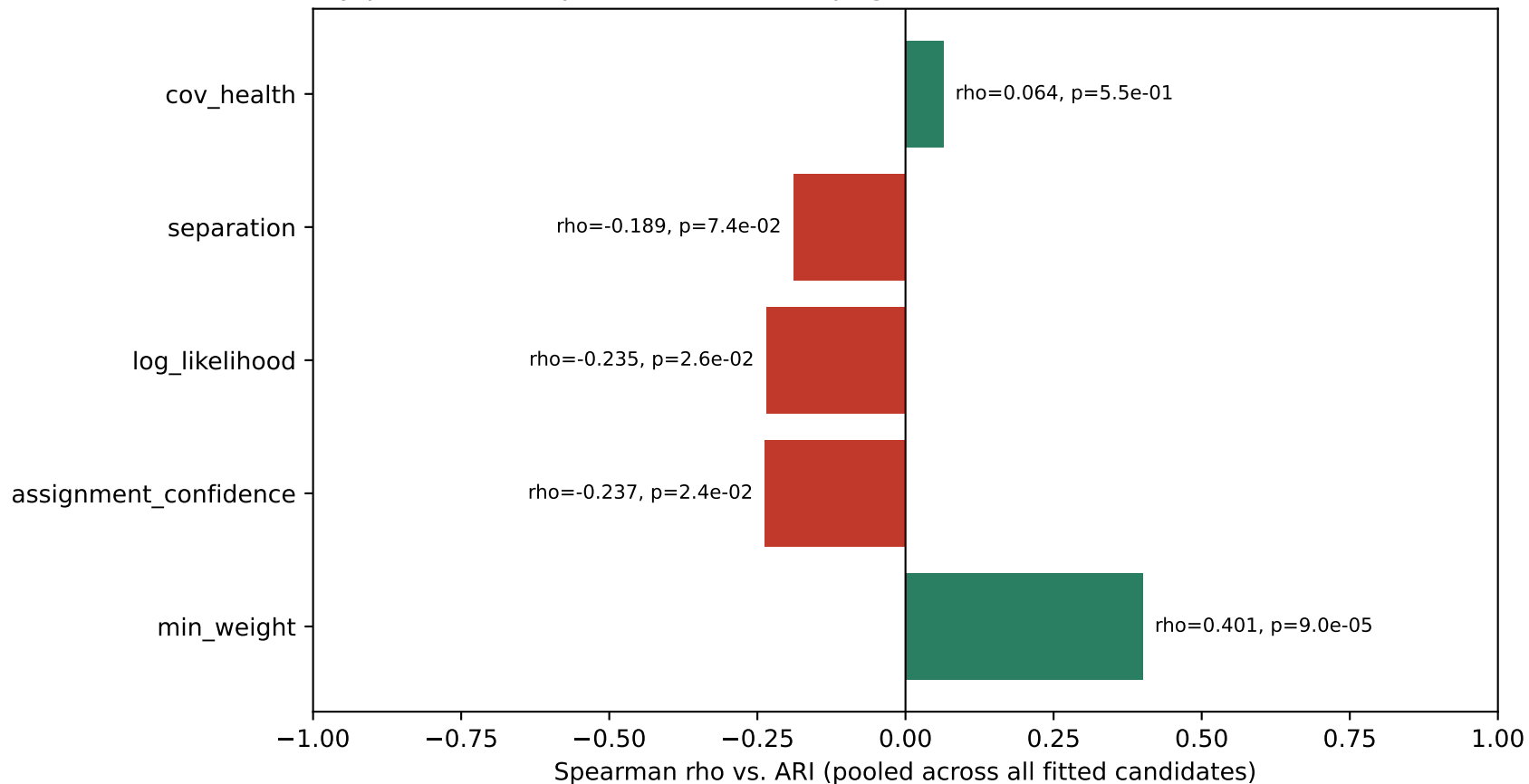
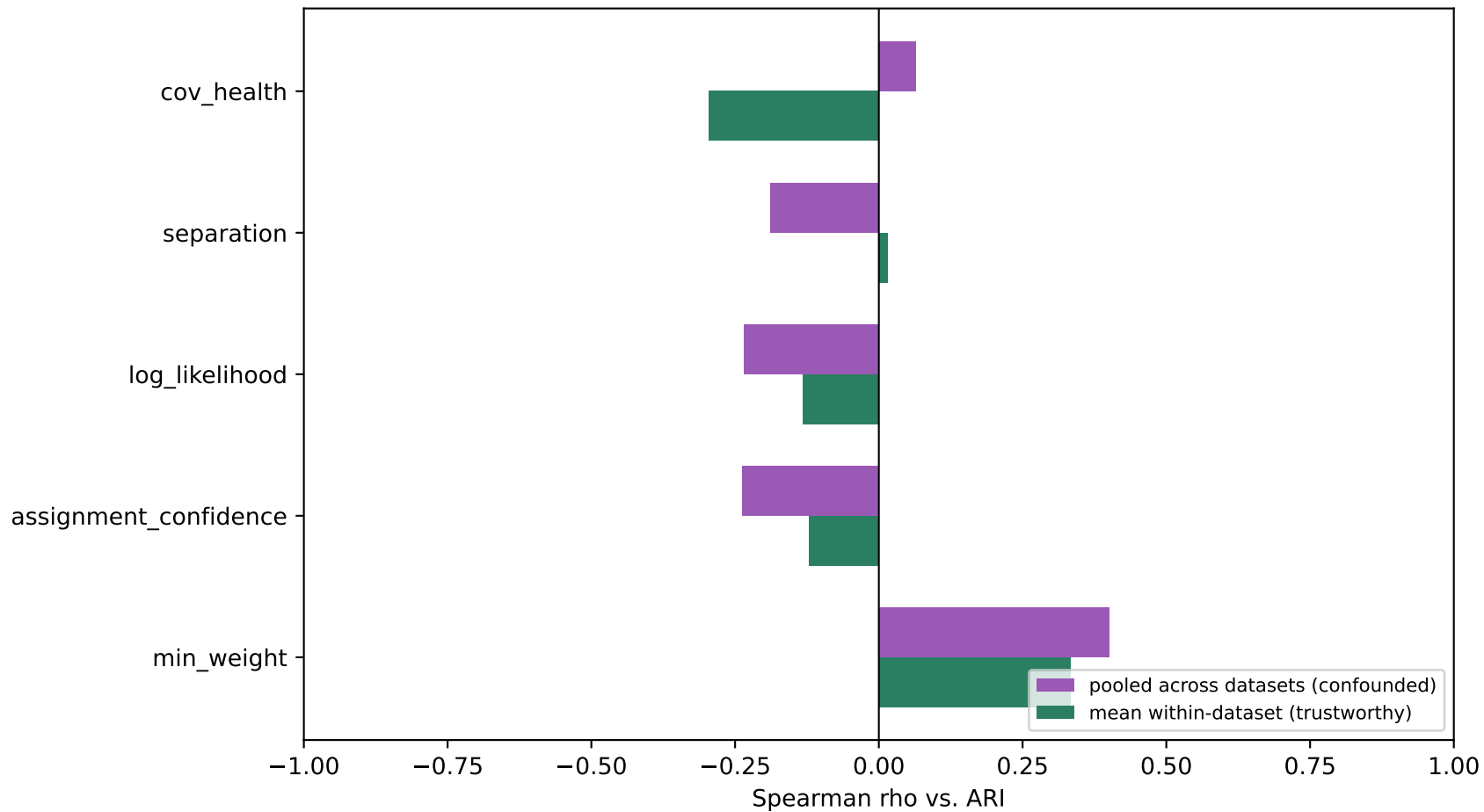


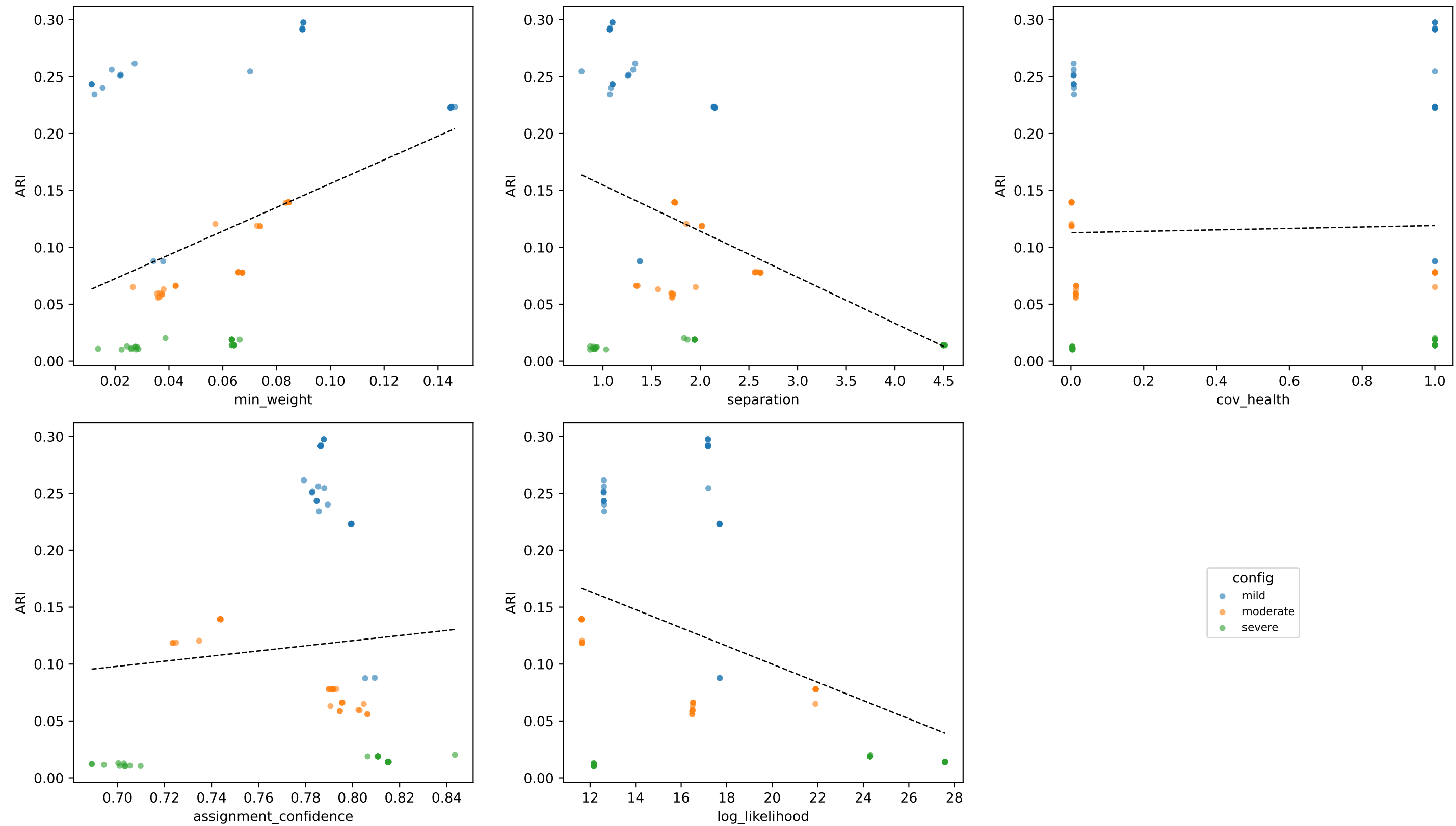
Which raw metrics actually predict ARI? (pooled -- see next page: this inflates/deflates metrics via cross-data)



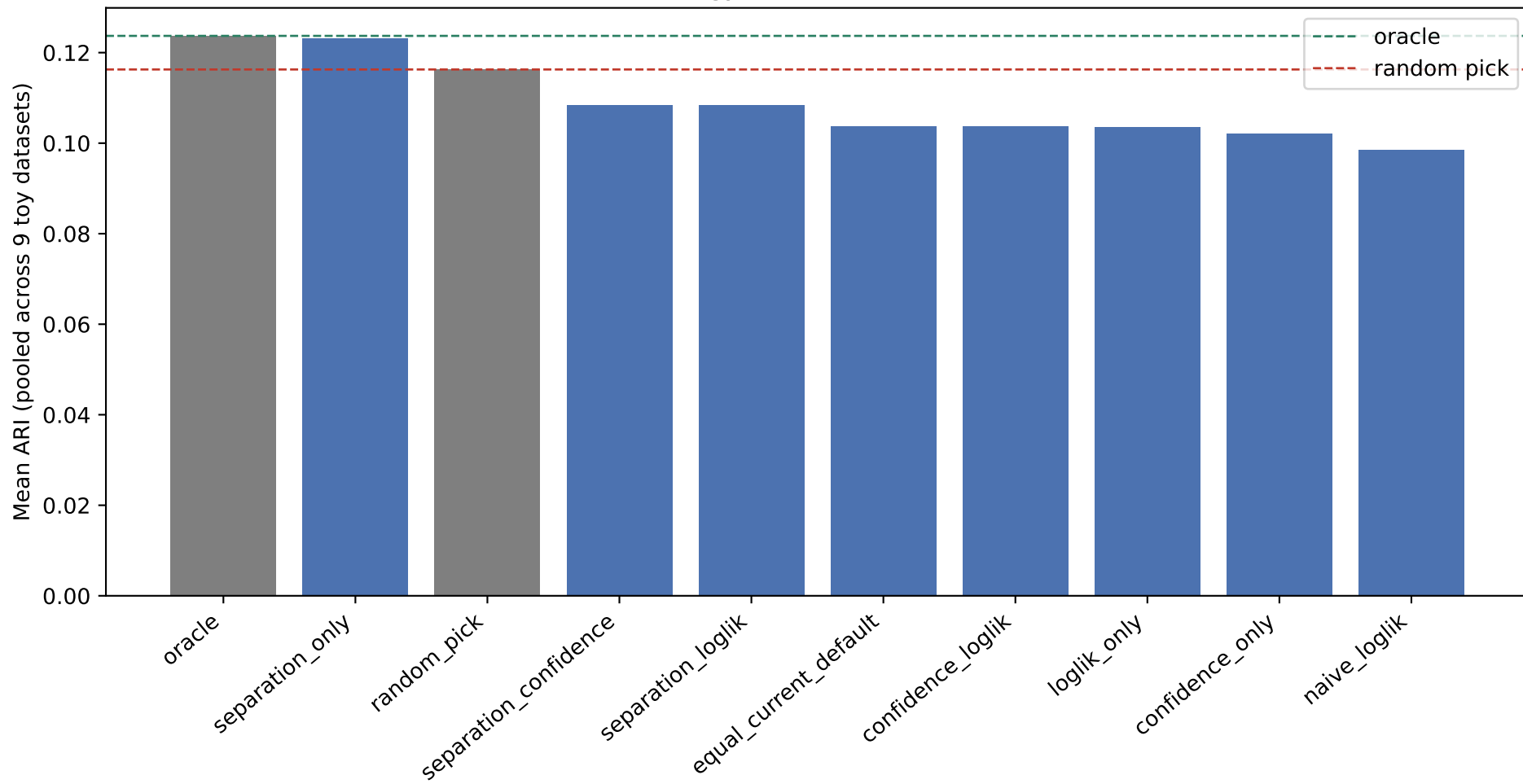
Pooled correlation can overstate (or invert) predictive power --
within-dataset rho is the number to trust



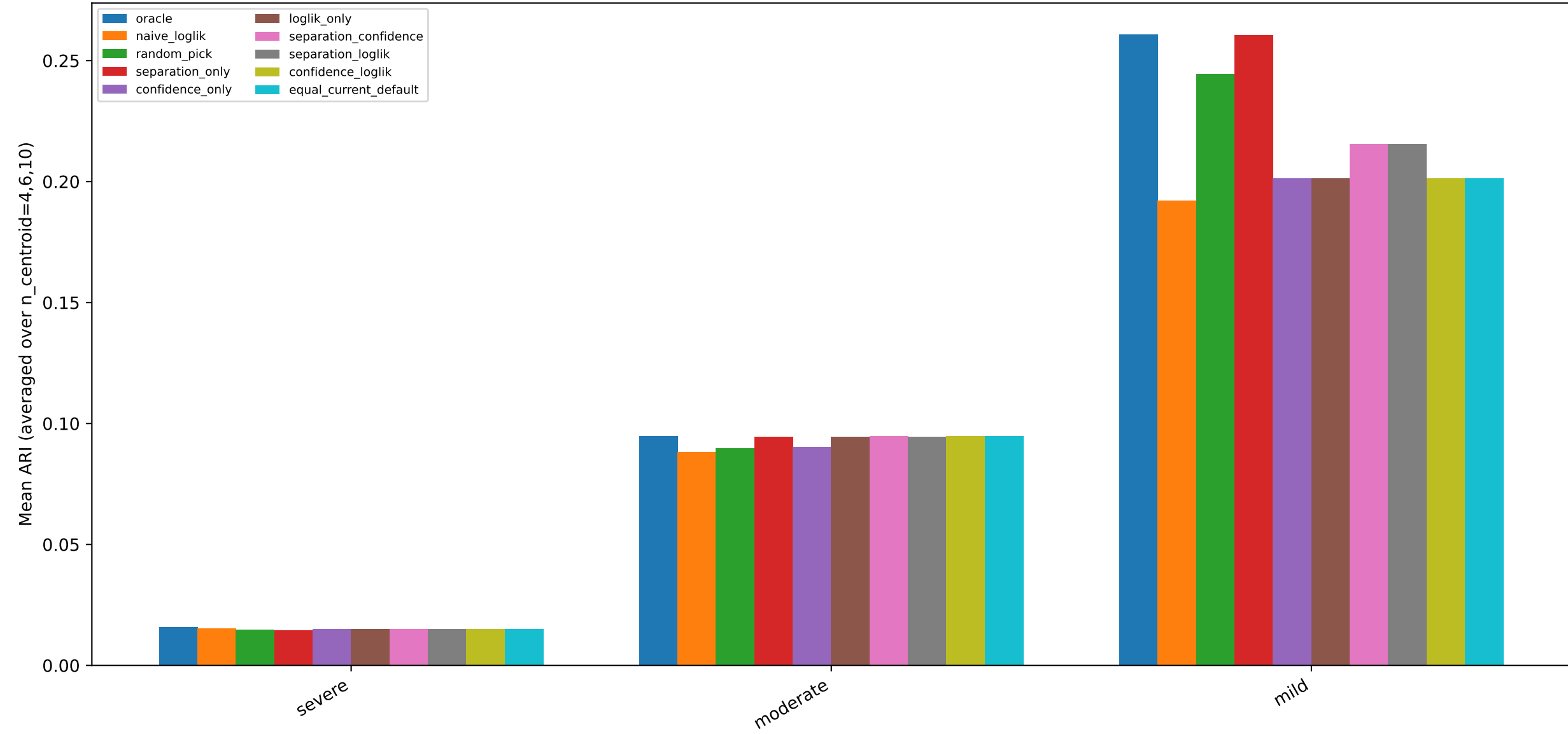
Metric value vs. ARI, per candidate (90 candidates, colored by stress config)



Selection strategy vs. oracle / random baselines



Strategy ARI by stress config — where does the metric choice matter?



Metric selection summary

Metric → ARI correlation ranking (pooled, $|\rho|$ descending):

min_weight	$\rho=+0.401$	$p=9.0e-05$
assignment_confidence	$\rho=-0.237$	$p=2.4e-02$
log_likelihood	$\rho=-0.235$	$p=2.6e-02$
separation	$\rho=-0.189$	$p=7.4e-02$
cov_health	$\rho=+0.064$	$p=5.5e-01$

Metric → ARI correlation ranking (WITHIN-DATASET mean, $|\rho|$ descending -- trust this one):

min_weight	mean_rho=+0.333	(9/9 datasets)
cov_health	mean_rho=-0.296	(4/9 datasets)
log_likelihood	mean_rho=-0.133	(9/9 datasets)
assignment_confidence	mean_rho=-0.122	(9/9 datasets)
separation	mean_rho=+0.015	(9/9 datasets)

Top 3 selection strategies by pooled mean ARI:

oracle	mean ARI=0.1237	
separation_only	mean ARI=0.1232	
random_pick	mean ARI=0.1163	
oracle	mean ARI=0.1237	(upper bound)
random_pick	mean ARI=0.1163	(lower bound)

AE pretraining loss vs. epoch, per dataset

